```
.686
.model flat
public _create_last
extern _VirtualAlloc@16
.code
_create_last proc
    push ebp
    mov ebp, esp
    push ebx
    push esi
    push edi

    mov esi, [ebp+8]
    xor ecx, ecx

count_unique:
    mov al, [esi]
    test al, al
    jz end_count
    inc ecx
    inc esi
    jmp count_unique

end_count:
    shl ecx, 2
    add ecx, 4

    push 0
    push 0
    push ecx
    push 0

    call _VirtualAlloc@16
    mov edi, eax

    mov esi, [ebp+12]
    xor ebx, ebx

fill_last:
    mov al, [esi]
    test al, al
    jz end_fill
    mov ecx, edi

check_existing:
    cmp word ptr [ecx], 0
    je add_new_entry
    cmp al, [ecx]
    je update_entry
    add ecx, 4
    jmp check_existing

update_entry:
    mov [ecx+2], bx
    jmp next_char
add_new_entry:
    mov [ecx], al
    mov [ecx+2], bx
next_char:
    inc esi
    inc ebx
    jmp fill_last
end_fill:
    mov dword ptr [ecx+4], 0
    mov eax, edi
    pop edi
    pop esi
    pop ebx
    pop ebp
    ret
_create_last endp

END
```